

TOPS Technologies

Server Security and Automation

Assessment A4 — Complete Answers

Theme: Startup Web Server Deployment

Total Time: 3 Hours

Assessment 4

Section A: Scenario-Based Concept Application

1. Explain the operational factors that justify selecting a specific Linux distribution for a startup's internal tool hosting environment.

When selecting a Linux distribution for a startup, the following factors should be considered:

- Stability: Stable distributions like RHEL and Ubuntu LTS provide reliable performance.
- Security: Regular security patches and SELinux/AppArmor protection.
- Long-Term Support (LTS): Reduces maintenance efforts.
- Package Management: Easy installation and updates using DNF/YUM (RHEL) or APT (Ubuntu).
- Community & Vendor Support: Large community and official support help solve issues quickly.
- Performance: Linux efficiently utilizes hardware resources.
- Compatibility: Supports Apache, Nginx, MySQL, Docker, Kubernetes, etc.
- Scalability: Easily supports future business growth.
- Cost: Open-source distributions reduce licensing costs.

2. Explain the implementation of SSH Key-Based Authentication to satisfy a security requirement where password logins must be disabled.

Steps:

Client

- `Ssh-keygen -t rsa -b 4096`

Copy the public key

- `cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`

Edit SSH configuration

- `sudo vi /etc/ssh/sshd_config`

Change

- `PasswordAuthentication no`
- `PubkeyAuthentication yes`
- `PermitRootLogin no`

Restart SSH

- `sudo systemctl restart sshd`

Verify

- `ssh user@server-ip`

3. Explain the configuration of Linux Permissions and Ownership for `/var/www`

Apache stores website files inside

- /var/www/html

Create group

- sudo groupadd webadmins

Add user

- sudo usermod -aG webadmins developer

Change ownership

- sudo chown -R root:webadmins /var/www

Set permissions

Directories

- sudo find /var/www -type d -exec chmod 775 {} \;

Files

- sudo find /var/www -type f -exec chmod 664 {} \;

Verify

- ls -ld /var/www
- ls -l /var/www/html

4. Explain how Logical Volume Management (LVM) facilitates dynamic storage expansion.

LVM allows storage to be resized without reinstalling Linux.

Components

- Physical Volume (PV)
- Volume Group (VG)
- Logical Volume (LV)

5. Explain the troubleshooting process for diagnosing SELinux blocking Apache.

Check SELinux status

- getenforce

Check audit logs

- sudo cat /var/log/audit/audit.log

View Apache logs

- sudo journalctl -xe

Restore security context

- sudo restorecon -Rv /var/www/html

Allow HTTP service

- sudo setsebool -P httpd_can_network_connect on

Test

- systemctl restart httpd

Open browser

- `http://server-ip`

6. Explain the safe configuration of Cron Jobs for automatic daily backup.

Cron automates repetitive tasks.

Edit cron

- `crontab -e`

Add

- `0 2 * * * tar -czf /backup/web_backup_$(date +%F).tar.gz /var/www`

Verify

- `crontab -l`

Check logs

- `grep CRON /var/log/cron`

Section B: Practical Application

1. System Identity

Task: Set the system hostname to startup-web-01 and verify the change.

Step 1: Check Current Hostname

- `hostnamectl`

Step 2: Set New Hostname

- `sudo hostnamectl set-hostname startup-web-01`

Step 3: Verify

- `hostname`

2. Network Setup

Task: Configure a Static IP Address on the Primary Interface and Restart the Network.

Step 1: Find Interface Name

- `ip addr`

Step 2: Configure Static IP

- `sudo nmcli con mod ens160 \`
- `ipv4.addresses 192.168.1.100/24 \`
- `ipv4.gateway 192.168.1.1 \`
- `ipv4.dns 8.8.8.8 \`
- `ipv4.method manual`

Replace ens160 with your interface name.

Step 3: Restart Network

- `sudo systemctl restart NetworkManager`

Step 4: Verify

- `ip addr`
- `ip route`
- `ping -c 4 8.8.8.8`

3. Secure Access

Task: Enable SSH Service and Start Automatically on Boot

Check SSH Status

- `systemctl status sshd`

Start SSH

- `sudo systemctl start sshd`

Enable at Boot

- `sudo systemctl enable sshd`

Verify

- `systemctl is-enabled sshd`

4. Access Control

Task: Create a Group named webadmins and User dev-ops with Restricted sudo Privileges.

Create Group

- `sudo groupadd webadmins`

Create User

- `sudo useradd dev-ops`

Set Password

- `sudo passwd dev-ops`

Add User to Group

- `sudo usermod -aG webadmins dev-ops`

Give Limited sudo Access

Open sudoers file

- `sudo visudo`

Add

- dev-ops ALL=(ALL) /usr/bin/systemctl, /usr/bin/dnf

This allows only these commands with sudo:

- systemctl
- dnf

Verify

- id dev-ops

5. Web Hosting

Task: Install Apache and Deploy a Custom index.html.

Install Apache

- sudo dnf install httpd -y

Start Apache

- sudo systemctl start httpd

Enable at Boot

- sudo systemctl enable httpd

Create Web Page

- sudo vi /var/www/html/index.html

Add

```
<!DOCTYPE html>
<html>
<head>
<title>TOPS Assessment</title>
</head>
<body>
<h1>Welcome to Startup Web Server</h1>
<p>Apache Server Running Successfully</p>
</body>
</html>
```

Verify

- http://server-ip

6. Hardening

Task: Configure Firewall to Allow Only Ports 22 and 80.

Start Firewalld

- `sudo systemctl enable --now firewalld`

Allow SSH

- `sudo firewall-cmd --permanent --add-service=ssh`

Allow HTTP

- `sudo firewall-cmd --permanent --add-service=http`

Reload Firewall

- `sudo firewall-cmd --reload`

Verify

- `sudo firewall-cmd --list-all`

7. Automation

Task: Create a Cron Job that Runs Daily at 02:00 AM to Compress /var/www into a .tar.gz Archive.

Create Backup Directory

- `sudo mkdir -p /backup`

Edit Cron

- `crontab -e`

Add

- `0 2 * * * tar -czf /backup/www_backup_$(date +%F).tar.gz /var/www`

Save and Exit

Verify Cron

- `crontab -l`

Test Backup Manually

- `sudo tar -czf /backup/test_backup.tar.gz /var/www`

Verify Backup

- `ls -lh /backup`

Section C: Integrated Case Study

Scenario

CloudScale, a boutique hosting provider, performed a kernel update on their primary web server. After reboot:

- The web application is unreachable (502 Gateway Error).
- SSH login takes more than 60 seconds.
- The server is online, but services are not functioning correctly, causing business downtime.

1. Diagnosis

Question:

Which log files and system commands would you use to identify why the Apache service failed to start?

Step 1: Check Apache Service Status

- `sudo systemctl status httpd`

Step 2: Check System Logs

- `sudo journalctl -u httpd`

Step 3: Check Apache Error Logs

- `sudo cat /var/log/httpd/error_log`

Step 4: Test Apache Configuration

- `sudo httpd -t`

Step 5: Check SELinux

- `getenforce`
- `sudo ausearch -m AVC`

Step 6: Verify Port 80

- `sudo ss -tulpn | grep :8`

Step 7: Check Disk Space

- `df -h`

A full filesystem can prevent Apache from starting.

2. Connectivity

Question:

Analyze why SSH login is experiencing high latency and suggest a fix related to DNS or authentication settings.

Possible Causes

Reverse DNS Lookup

SSH attempts to resolve the client's hostname before login.

Disable DNS lookup.

Edit:

- `sudo vi /etc/ssh/sshd_config`

Add or modify:

- `UseDNS no`

Restart SSH:

- `sudo systemctl restart sshd`

GSSAPI Authentication

Disable unnecessary Kerberos authentication.

In `/etc/ssh/sshd_config`:

- `GSSAPIAuthentication no`

Restart:

- `sudo systemctl restart sshd`

Check DNS

- `nslookup google.com`

Verify DNS configuration:

- `cat /etc/resolv.conf`

Verify Network

- `ping -c 4 8.8.8.8`

Verify SSH

- `time ssh localhost`

Login should be much faster.

3. Recovery

Question:

Outline the step-by-step CLI process to manually restart the failed services and verify their operational status.

Restart Apache

- `sudo systemctl restart httpd`

Check Status

- `sudo systemctl status httpd`

Restart SSH

- `sudo systemctl restart sshd`

Check Status

- `sudo systemctl status sshd`

Enable Services at Boot

- `sudo systemctl enable httpd`
- `sudo systemctl enable sshd`

Verify Listening Ports

- `sudo ss -tulpn`

Verify Website

- `curl http://server-ip`

Verify SSH

- `ssh localhost`

Check Overall System

- `systemctl --failed`

This lists any failed services.

4. Preventive Action

Question:

Propose a configuration change to ensure the web server automatically attempts to restart if it encounters a runtime failure.

Create a systemd override.

- `sudo systemctl edit httpd`

Add:

- [Service]
- Restart=always
- RestartSec=5

Save and exit.

Reload systemd:

- `sudo systemctl daemon-reload`

Restart Apache:

- `sudo systemctl restart httpd`

Verify:

- `systemctl show httpd | grep Restart`

5. Security

Question:

Justify the use of SSH Key-Based Authentication over passwords to prevent brute-force attacks during this vulnerable state.

SSH Key-Based Authentication uses a public/private key pair instead of passwords.

Advantages

- Passwords are not transmitted over the network.
- Prevents brute-force and dictionary attacks.
- More secure than password authentication.
- Supports automated scripts without exposing passwords.
- Easy to disable a compromised key by removing it from `authorized_keys`.
- Can be combined with disabled root login and MFA for stronger security.
- Provides better security for remote server administration.

Recommended SSH Configuration

Edit:

- `sudo vi /etc/ssh/sshd_config`

Set:

- PermitRootLogin no
- PasswordAuthentication no
- PubkeyAuthentication yes
- UseDNS no
- GSSAPIAuthentication no

Restart SSH:

- `sudo systemctl restart sshd`